

Hanif Carroll

Applied AI Engineer | AI tools for business workflows

U.S. citizen · Buenos Aires, Argentina · U.S. hours · Open to remote employment or contracts through HC Studio LLC

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Summary

Applied AI Engineer with eight years of software engineering experience, building reliable AI products systems around business workflows. Recent client work spans AI image editing, language-learning pipelines, billing systems, and recoverable background processing. Helped Genrupt support its first 200 paying customers and took Palabruno from an early prototype to launch across iOS, Android, and web.

Technical Skills

Backend & Data: TypeScript, Node.js, Python, FastAPI, Pydantic, PostgreSQL, SQL, Cloudflare D1

Applied AI: RAG / hybrid retrieval, NLP, OCR / inpainting, structured outputs, tool calling / MCP, LangGraph, evaluations / LangSmith

Frontend & Mobile: React, Next.js, SvelteKit, React Native / Expo, TanStack Start, Playwright, Storybook, Vitest

Infrastructure & Payments: Cloudflare Workers / Workflows / Containers, Docker, Stripe, RevenueCat

AWS: Amazon Bedrock, Lambda, DynamoDB, SQS, Step Functions, SAM, CloudFormation, IAM, CloudWatch

Work Experience

Founder & AI Product Engineer

HC Studio

Mar 2024 – Present

Build client AI products and business workflows from requirements through integrations, UX, and launch.

Genrupt — Contract product engineering

- Built an AI image editor that turns detected text into editable layers using OCR and inpainting; evaluated self-hosted models on RunPod and selected providers for latency and removal quality.
- Prevented concurrent AI generation from overdrawing account balances through atomic PostgreSQL credit reservations, row locking, and failure cleanup.
- Moved image generation to pg-boss jobs; made video retries reuse provider jobs and settle costs idempotently.
- Exposed market-analysis and Amazon listing workflows through MCP, with OAuth, organization-aware access, and asynchronous results.

Palabruno — Contract full-stack engineering

- Shipped iOS, Android, and web apps with Supabase, RevenueCat subscriptions, and Stripe billing.
- Engineered a Spanish reader compiler combining NLP, linguistic rules, and constrained LLM generation to produce tappable words, phrases, and contextual learning cards, with targeted repair and regression evaluations.
- Implemented durable text-processing workflows with persisted progress, failure recovery, and per-stage latency and model-cost tracking.

Senior Frontend Engineer

LeagueApps, later via Nearsure

Nov 2021 – Feb 2024

- Modernized tournament scheduling with React on the company's first migration team, helping establish shared frontend patterns and components.
- Built drag-and-drop scheduling, conflict detection, and double-booking prevention, reducing organizer scheduling time by 45%.
- Built self-service reporting with AG Grid; used virtualization and memoization for tens of thousands of rows, reducing load and interaction lag by 70%.

Software Engineer

Capgemini

Jan 2019 – Oct 2021

- Maintained five Java Spring Boot microservices for GE Aviation, collaborating across offshore teams.
- Delivered a reporting module for Disney Cruise Line's ship and crew management system and executive KPI dashboards for Cisco.
- Reduced automated test-suite runtime by 60%.

Frontend Engineer

8base

Nov 2017 – Nov 2018

- Built Jira / GitLab CI/CD automation and refactored 20+ shared UI components into a reusable library, reducing duplicate code by 30%.

Selected Projects

Switchboard — AI change-management demo | [Live demo](#)

- Built a Python/FastAPI and LangGraph agent that investigates customer change requests, cites supporting records and policies, and requires human approval before execution.
- Implemented persistent workflows on AWS with permission and configuration checks, duplicate-request handling, and signed test events to verify delivery.

Casamo — Independent product | [casamo.app](#)

- Built an AI stay-research product that turns Airbnb and Booking.com listings, reviews, and photos into cited shortlists, validating evidence and flagging unverified requirements.
- Built resumable AI investigations with Cloudflare Workflows, D1, and R2, preserving evidence across failures and restoring credits idempotently.

BA Eventos — Independent product | [baeventos.com](#)

- Built an AI event concierge that combines semantic and keyword search with reranking to recommend relevant events from the catalog.

Languages

- **English** — Native
- **Spanish** — Professional working proficiency

Education

- **B.S. in Health Sciences**, University of South Florida